

Gate E7 report — run_v1.21_r1 (Paper IV v1.21)

Verdict: **PASS**. Evidence: 20_EVIDENCE/E7/.

E7 — Attribution revalidation

Sources: cited versions saved in run_v1.2_r1 (20_EVIDENCE/E7/sources/, hashes in its SOURCES_SHA256.txt; the lead auditor re-read [15] v1 pp.1–3, 8–11, 17–20, 23 and the [5] README at the cited commit). Later-version check on 2026-09-30 23:07 UTC: arXiv 2609.20871 abs page lists only [v1] (sha256 e1ce80ea..., identical to the page saved on 2026-09-30 12:28 UTC); arXiv 1902.06135 only [v1]; N0zoM1z0/erdos-81 latest commits cbde8a0a (2026-09-08), 2dd0615a, fc7712b1 — HEAD unchanged. Saved in sources/.

ID	v1.2 issue	v1.21 text	Check against source	Resolution
X-06	Table 4 / 1.1257 described [15]'s $K_3 \dots K_L$ use as a far-transfer branch paying the budget	Table 4 row: " $K_3 \dots K_L$ in the approximation used for localization; q_L in Lemma 3.4; cp in Theorem 1.1; $\Delta_{\{3..L\}}$ for the auxiliary model"; §8.1: conditional calculation "is not a description of the final construction in [15, Theorem 1.1]: that proof uses Lemma 3.4 through structural localization, then a minimum-counterexample argument and the triangle constructions of [15, §§4–5]"	[15] p.10–11 (Lemma 3.4 inside Thm 1.3 localization; (1.5) with $L = 4$), p.19–20 (minimal counterexample $K \rightarrow \infty$ via Prop 5.4; (5.11) from §4 triangle constructions)	RESOLVED
X-07	"stronger" c_4 hypothesis	"under the weaker near-extremality hypothesis on c_4 : the corresponding lower bound on cp implies it, since $cp \leq c_4$ "	Logic: $cp(G) \geq Q - \delta \Rightarrow c_4(G) \geq Q - \delta$	RESOLVED
X-08	"order-four version of (1.5)"	"recovers its approximation (1.5) with a conclusion stated for $c_4 \dots$ The proof of (1.5) in [15] already chooses $L = 4$; that choice is not a new ingredient here"	[15] p.11 "For (1.5), use $L = 4$ in Lemmas 3.1 and 3.4"	RESOLVED
X-09	Abstract/Thm A omitted [15, Cor 1.2]	Abstract: "also obtained in [5] and [15, Corollary 1.2]"; Theorem A remark likewise	[15] Cor 1.2 (p.2) and abstract (p.1)	RESOLVED
X-10	[5] authorship / Jacobian	Ref [5] keeps printed "Anonymous" and adds the README credits (Morluto proof; N0zoM1z0 manuscript and Lean; Jacobian at Preference Labs); AI/tools section: "Jacobian is also credited by the project behind [5]"	README at cbde8a0a, "## Credits": proof by Morluto with GPT assistance; Jacobian "a research tool developed at Preference Labs, played a substantial role"; manuscript and Lean by N0zoM1z0; "a Preference Labs research project"	RESOLVED (OBSERVATION: the manuscript writes "PreferenceLabs"; the README writes "Preference Labs"; the README also says the proof had GPT assistance — not required to be repeated)
X-11	"reduced from 10^{32} to $4 \cdot 10^{12n}$ ambiguous	"During the development of our near constructor ... This compares versions of that constructor, not the global thresholds of different papers."	—	RESOLVED
[15] Theorem C attribution (historical r2 F-01)	—	Unchanged from v1.2 (credited)	—	Remains resolved
Localization vs final construction	—	§8.1 as above	[15] pp.9–11, 17–20	Accurate
$cp \leq c_4$ direction	—	§8.1 (X-07)	—	Accurate
Prior use of $L = 4$	—	§8.1 (X-08)	—	Accurate
[23] and the classification wrappers	—	[23] note now names the unrestricted-classification wrappers	AuditorASProbe.log	Accurate (see E3, X-27)

No inference of bibliographic originality is drawn from dependency cones. Coverage limits as in run_v1.2_r1 (no MathSciNet/zbMATH/Scholar; not a complete novelty search).

E7 verdict: PASS (X-06–X-11 resolved; one OBSERVATION on spelling "PreferenceLabs").